TSM 099 **61797**

Arithmetic Test - Practice Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate at (a) x = 4 and (b) x = -4. $2x^2 - 2x + 1$	1. (a) 10 pts (b) 10 pts
2. Simplify the square root. $\sqrt{147}$	2. 20 pts
3. Evaluate each side at x = 1, -3, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2 - 36}{x - 6} = \frac{x^2 + 13x + 42}{x + 7}$	3. x = 1 4 pts x = -3 4 pts x = 0 4 pts Conclusion 8 pts
4. Reduce each fraction to lowest terms. (a) $\frac{18}{24}$ (b) $\frac{10}{14}$	4. (a) 10 pts (b) 10 pts
5. Evaluate the principal square root in each part. (a) $\sqrt{49}$ (b) $\sqrt{-25}$	5. (a) 10 pts (b) 10 pts

61797

TEST PAGE 1

1. (a)

25 — 10 pts

(b)

41 — 10 pts

2.

$7\sqrt{3}$ — 20 pts

3. x = 1

Left 7; right 7; yes — 4 pts

x = -3

Left 3; right 3; yes — 4 pts

x = 0

Left 6; right 6; yes — 4 pts

Conclusion

All tested values work; this alone does not prove an identity. — 8 pts

4. (a)

$\frac{3}{4}$ — 10 pts

(b)

$\frac{5}{7}$ — 10 pts

5. (a)

7 — 10 pts

(b)

$5i$ — 10 pts

TSM 099 **84507**

Arithmetic Test - Practice Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate the principal square root in each part. $(a)\quad \sqrt{-25}$ $(b)\quad \sqrt{64}$	1. (a) 10 pts (b) 10 pts
2. Evaluate each side at x = 2, -4, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-64}{x-8} = \frac{2x^2+20x+48}{x+6}$	2. x = 2 4 pts x = -4 4 pts x = 0 4 pts Conclusion 8 pts
3. Evaluate at (a) x = 3 and (b) x = -3. $(x-2)(x+4)$	3. (a) 10 pts (b) 10 pts
4. Reduce each fraction to lowest terms. $(a)\quad \frac{25}{40}$ $(b)\quad \frac{60}{75}$	4. (a) 10 pts (b) 10 pts
5. Simplify the square root. $\sqrt{18}$	5. 20 pts

84507

TEST PAGE 1

1. (a)

$5i$ — 10 pts

(b)

8 — 10 pts

2. x = 2

Left 10; right 12; no — 4 pts

x = -4

Left 4; right 0; no — 4 pts

x = 0

Left 8; right 8; yes — 4 pts

Conclusion

Only the listed matching value works; the equation is not an identity. — 8 pts

3. (a)

7 — 10 pts

(b)

-5 — 10 pts

4. (a)

$\frac{5}{8}$ — 10 pts

(b)

$\frac{4}{5}$ — 10 pts

5.

$3\sqrt{2}$

20 pts

TSM 099 **22003**

Arithmetic Test - Practice Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Simplify the square root. $\sqrt{18}$	1. 20 pts
2. Reduce each fraction to lowest terms. (a) $\frac{105}{135}$ (b) $\frac{35}{56}$	2. (a) 10 pts (b) 10 pts
3. Evaluate the principal square root in each part. (a) $\sqrt{144}$ (b) $\sqrt{-121}$	3. (a) 10 pts (b) 10 pts
4. Evaluate both values in the formula for a = 2, b = -5, c = 2. $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	4. 20 pts
5. Evaluate each side at x = 3, -1, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2 - 49}{x - 7} = \frac{2x^2 + 8x - 24}{x + 6}$	5. x = 3 4 pts x = -1 4 pts x = 0 4 pts Conclusion 8 pts

22003

TEST PAGE 1

1.

$3\sqrt{2}$

20 pts

2. (a)

$\frac{7}{9}$

10 pts

(b)

$\frac{5}{8}$

10 pts

3. (a)

12

10 pts

(b)

$11i$

10 pts

4.

$x = \frac{1}{2}, 2$

20 pts

5. x = 3

Left 10; right 2; no

4 pts

x = -1

Left 6; right -6; no

4 pts

x = 0

Left 7; right -4; no

4 pts

Conclusion

None of these values works; this alone does not prove no solution.

8 pts

TSM 099 **64765**

Arithmetic Test - Practice Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate at (a) x = 2 and (b) x = -2. $x^2 - 2x + 2$	1. (a) 10 pts (b) 10 pts
2. Evaluate the principal square root in each part. $(a)\quad \sqrt{-144}$ $(b)\quad \sqrt{64}$	2. (a) 10 pts (b) 10 pts
3. Reduce each fraction to lowest terms. $(a)\quad \frac{70}{112}$ $(b)\quad \frac{6}{10}$	3. (a) 10 pts (b) 10 pts
4. Evaluate each side at x = 4, -3, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-25}{x-5} = \frac{x^2+13x+40}{x+8}$	4. x = 4 4 pts x = -3 4 pts x = 0 4 pts Conclusion 8 pts
5. Simplify the square root. $\sqrt{98}$	5. 20 pts

64765

TEST PAGE 1

1. (a)

2 — 10 pts

(b)

10 — 10 pts

2. (a)

$12i$ — 10 pts

(b)

8 — 10 pts

3. (a)

$\frac{5}{8}$ — 10 pts

(b)

$\frac{3}{5}$ — 10 pts

4. x = 4

Left 9; right 9; yes — 4 pts

x = -3

Left 2; right 2; yes — 4 pts

x = 0

Left 5; right 5; yes — 4 pts

Conclusion

All tested values work; this alone does not prove an identity. — 8 pts

5.

$7\sqrt{2}$

20 pts

TSM 099 **99821**

Arithmetic Test - Practice Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Evaluate each side at x = 1, -2, 0. In each response row record the left value, right value, and equality decision. Then state what the tested values establish; do not solve the equation. $\frac{x^2-49}{x-7} = \frac{2x^2+23x+56}{x+8}$	1. x = 1 4 pts x = -2 4 pts x = 0 4 pts Conclusion 8 pts
2. Reduce each fraction to lowest terms. (a) $\frac{35}{49}$ (b) $\frac{30}{40}$	2. (a) 10 pts (b) 10 pts
3. Simplify the square root. $\sqrt{75}$	3. 20 pts
4. Evaluate at (a) x = 2 and (b) x = -2. $(x-1)(x+2)$	4. (a) 10 pts (b) 10 pts
5. Evaluate the principal square root in each part. (a) $\sqrt{-121}$ (b) $\sqrt{36}$	5. (a) 10 pts (b) 10 pts

99821

TEST PAGE 1

1. x = 1

Left 8; right 9; no — 4 pts

x = -2

Left 5; right 3; no — 4 pts

x = 0

Left 7; right 7; yes — 4 pts

Conclusion

Only the listed matching value works; the equation is not an identity. — 8 pts

2. (a)

$\frac{5}{7}$ — 10 pts

(b)

$\frac{3}{4}$ — 10 pts

3.

$5\sqrt{3}$ — 20 pts

4. (a)

4 — 10 pts

(b)

0 — 10 pts

5. (a)

$11i$ — 10 pts

(b)

6 — 10 pts